

CONTENTS

TECHNOLOGY

- 21 **FUTURE:** Importance Of Drones And Types Of Drones For Wildlife Monitoring
- 26 **CAREER:** Why And How To Become An Open Source Contributor
- 30 **TRENDS:** Quantum Key Distribution: The Future Of Secure Communication
- 34 **TRENDS:** Enter The Metaverse For Better Or For Worse

DESIGN

- 40 **TRENDS:** The Evolution Of LCD Displays And How It Matters
- 44 **CIRCUIT:** Why Transition From NOR To NAND Flash Memory
- 48 **CIRCUIT:** Analogue Front Ends For Heart Rate Monitors

DO-IT-YOURSELF

- 54 **PROJECT:** Study Of Rectified Waveforms Using Arduino Uno
- 56 **PROJECT:** Design Of Interrupt Based Matrix Keypad Embedded-C Driver For A Microcontroller
- 60 **PROJECT:** Smart Switch Box For Electric Iron
- 64 **CIRCUIT IDEA:** Simple Two-Way Intercom
- 66 **PROJECT:** Data Logging Using microSD And Raspberry Pi Pico
- 70 **PROJECT:** AI Camera For Tracking Suspicious Activities

STRATEGY

- 76  **GROWTH:** India's EV Industry Success Depends On How Soon It Stops Seeing It From A Global Lens

- 80 **GROWTH:** Why Products For Global Markets Need To Meet Standards



- 85 **GROWTH:** Semiconductor & Display Fabs: Devil Is In The Execution, Success In Collaboration



- 91 **POLICY:** Union Budget 2022-23: What It Means For India's Tech And Electronics Industry



REGULARS

- 08 **From You:** Your Suggestions and Things You Wanted To Know!
- 10 **Research & Innovation Updates**
- 14 **Truly Innovative Electronics**
- 16 **New Products**
- 74 **Dashboard:** Key Industry Trends That Can Shape The Future Of Your Business
- 96 **Industry News:** Industry Updates

EDITOR : Rahul Chopra

EDITORIAL : Phone: +91-9811155335
CORRESPONDENCE E-mail: editop@efy.in

SUBSCRIPTIONS & MISSING ISSUES : Phone: +91-9811155335
 E-mail: support@efy.in

BACK ISSUES, BOOKS, KITS, COMPONENTS, etc : Phone: +91-9811155335
 E-mail: support@efy.in

NEWSSTAND DISTRIBUTION : Phone: +91-9811155335
 E-mail: efycirc@efy.in

ADVERTISEMENTS
NEW DELHI (HEAD OFFICE) : Phone: +91-9811155335
 E-mail: efyenq@efy.in

MUMBAI : E-mail: rmowest@efy.in

BENGALURU : E-mail: rmosouth@efy.in

CHINA : Worldwide Focus Media
 E-mail: china@efy.in

GERMANY : PMS Plantenberg Media Service GmbH
 E-mail: germany@efy.in

JAPAN : Tandem Inc.
 E-mail: japan@efy.in

TAIWAN : J.K. Media
 E-mail: taiwan@efy.in

UNITED KINGDOM : ASA Media
 E-mail: uk@efy.in

UNITED STATES : E & Tech Media
 E-mail: usa@efy.in

Printed, published and owned by Ramesh Chopra. Printed at International Print-o-Pack Ltd, C-4 to C-11, Hosliery Complex, Phase-II Extension, NOIDA-201305, Gautam Budh Nagar, Uttar Pradesh, on the first day of each month and published from D-87/1, Okhla Industrial Area, Phase-1, New Delhi 110020. Copyright 2022. All rights reserved throughout the world. Reproduction of any material from this magazine in any manner without the written permission of the publisher is prohibited. Although every effort is made to ensure accuracy, no responsibility whatsoever is taken for any loss due to publishing errors. EFY will not be responsible for any wrong claims made by an advertiser. Disputes, if any, will be settled in a New Delhi court only.

SUBSCRIPTION RATES

EFY = Electronics For You

PRINT MAGAZINE	1 YEAR (12 copies each)	3 YEARS (36 copies each)	5 YEARS (60 copies each)
WITHIN INDIA (IN RUPEES)			
EFY (Rs 100/copy)	840	2150	3000
SAARC COUNTRIES (IN US\$ BY AIR)			
EFY	50	135	NA
OTHER COUNTRIES (IN US\$ BY AIR)			
EFY	100	270	NA

Courier delivery within India @ Rs 50 per copy

e-magazine subscriptions within India are available at half the rates mentioned here.

Overseas rates for each e-magazine in US\$:
 1 year: \$12; 3 years: \$33; 5 years: \$50 only

Please send payments only in favour of **EFY Enterprises Pvt Ltd** through demand draft/cheque/money order to D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020, or online at: <http://subscribe.efyindia.com> or scan the QR code.

